

Revisions After the Proposal Defense

10 October 2026 · Alexandra Sulit, Willard Soriano, Steven Evian Lozano · Page numbers refer to the attached revised proposal (120 pages).

Response to the Panel

CONCERN	RESPONSE	WHERE
Chapter 1 has no Background of the Study.	The content was present but had no subsection headings. Chapter 1 is now divided into numbered subsections, beginning with 1.1 Background of the Study, as Chapters 2 and 3 are.	Ch. 1, pp. 1–14
How is this a computer science study rather than an IT project?	The core contribution is an algorithm: a static analysis that predicts what a command will change before it runs, so that what it destroys can be backed up and restored. It builds on established program-analysis theory and is evaluated independently of the interface.	pp. 24, 62, 74
The objectives are not aligned with Computer Science. (Sir Joel)	The general objective is now the algorithm; specific objectives 1–3 are its formal model, its design, and its evaluation. The title, background, scope, significance, and research gap were re-aligned to match. New title: <i>SynapseOS: Recoverability Analysis of Language-Model-Generated Shell Commands in a Conversational Interface Layer</i> .	§1.3, pp. 6–8
How much of the result comes from the algorithm, and how much from its rule tables? (adviser, earlier review)	Now measured. With the same rule table but the algorithm’s reasoning disabled, data recovery falls from 89.5% to 35.4% ($p = 2.1 \times 10^{-53}$). The algorithm, not the table, drives the result.	§3.1(h), pp. 84–86

New Evidence at a Glance

- **The algorithm drives the result:** recovery drops from 89.5% to 35.4% without its reasoning (pp. 84–86).
- **Our baseline is not a weak one:** the safety checks in two open-source AI coding tools, built for code projects where git can undo mistakes rather than for personal files, let 73–97% of file-destroying commands through without asking, against 31% for our baseline and 0% for the algorithm (p. 78).
- **It works end to end:** run through the full system on real files, undo restored deleted files only with the algorithm enabled (p. 70).

Limitations We Disclose

- Undo currently reverts the most recent change regardless of folder; it will be scoped before data collection (p. 92).
- The small local language model sometimes mistranslates loosely worded requests, as already noted in the scope (p. 13); this does not affect the algorithm.

Supporting material (the comparison with open-source tools, a recorded end-to-end verification, and per-command results for all 777 test commands) is available on request.